SOLVED PRACTICE WORKBOOK

Nuclear Power and the Three-Stage Programme - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

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Nuclear Power and the Three-Stage Programme - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Fission-power chain?

A. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. B. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. C. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. D. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable.

Answer: A. Explanation: Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Fission-power chain?

A. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. B. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. C. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. D. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable.

Answer: B. Explanation: Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Fission-power chain without changing its institution, unit or status?

A. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. B. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. C. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. D. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast.

Answer: C. Explanation: Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Fission-power chain?

A. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. B. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. C. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. D. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation.

Answer: D. Explanation: Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Criticality boundary?

A. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. B. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. C. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. D. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable.

Answer: A. Explanation: Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Criticality boundary?

A. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. B. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. C. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. D. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel.

Answer: B. Explanation: Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Criticality boundary without changing its institution, unit or status?

A. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. B. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. C. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. D. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel.

Answer: C. Explanation: Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Criticality boundary?

A. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. B. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. C. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. D. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation.

Answer: D. Explanation: Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Control-safety boundary?

A. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. B. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. C. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. D. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy.

Answer: A. Explanation: Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Control-safety boundary?

A. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. B. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. C. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. D. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy.

Answer: B. Explanation: Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Control-safety boundary without changing its institution, unit or status?

A. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. B. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. C. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. D. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.

Answer: C. Explanation: Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission

rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Control-safety boundary?

A. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. B. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. C. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. D. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable.

Answer: D. Explanation: Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Thermal-fast boundary?

A. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. B. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. C. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. D. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy.

Answer: A. Explanation: Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Thermal-fast boundary?

A. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. B. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. C. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. D. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states.

Answer: B. Explanation: Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Thermal-fast boundary without changing its institution, unit or status?

A. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. B. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. C. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. D. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states.

Answer: C. Explanation: Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a

reactor thermal or fast. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Thermal-fast boundary?

A. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. B. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. C. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. D. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast.

Answer: D. Explanation: Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Fissile-fertile boundary?

A. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. B. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. C. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. D. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.

Answer: A. Explanation: U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Fissile-fertile boundary?

A. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. B. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. C. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. D. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.

Answer: B. Explanation: U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Fissile-fertile boundary without changing its institution, unit or status?

A. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. B. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. C. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. D. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.

Answer: C. Explanation: U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Fissile-fertile boundary?

A. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. B. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. C. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. D. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel.

Answer: D. Explanation: U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Stage-1 boundary?

A. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. B. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. C. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. D. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states.

Answer: A. Explanation: Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Stage-1 boundary?

A. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. B. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. C. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. D. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.

Answer: B. Explanation: Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Stage-1 boundary without changing its institution, unit or status?

A. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. B. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. C. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. D. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation.

Answer: C. Explanation: Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Stage-1 boundary?

A. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. B. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. C. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. D. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy.

Answer: D. Explanation: Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Plutonium-stream boundary?

A. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. B. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. C. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. D. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium.

Answer: A. Explanation: Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Plutonium-stream boundary?

A. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. B. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. C. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. D. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium.

Answer: B. Explanation: Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Plutonium-stream boundary without changing its institution, unit or status?

A. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. B. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. C. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. D. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop.

Answer: C. Explanation: Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. The other options change the scientific category, institutional owner, measurement, mission rung or

operating status.

Q28. Which option avoids the standard UPSC close-option trap about Plutonium-stream boundary?

A. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. B. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. C. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. D. Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states.

Answer: D. Explanation: Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Stage-2 boundary?

A. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. B. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. C. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. D. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation.

Answer: A. Explanation: Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Stage-2 boundary?

A. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. B. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. C. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. D. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium.

Answer: B. Explanation: Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Stage-2 boundary without changing its institution, unit or status?

A. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. B. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. C. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. D. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints.

Answer: C. Explanation: Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Stage-2 boundary?

A. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. B. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. C. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. D. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.

Answer: D. Explanation: Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies PFBR-status boundary?

A. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. B. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. C. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. D. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop.

Answer: A. Explanation: The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of PFBR-status boundary?

A. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. B. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. C. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. D. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints.

Answer: B. Explanation: The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses PFBR-status boundary without changing its institution, unit or status?

A. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. B. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. C. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. D. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged.

Answer: C. Explanation: The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. The other options change the scientific category, institutional owner, measurement,

mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about PFBR-status boundary?

A. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. B. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. C. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. D. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation.

Answer: D. Explanation: The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Stage-3 boundary?

A. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. B. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. C. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. D. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop.

Answer: A. Explanation: Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Stage-3 boundary?

A. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. B. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. C. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. D. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop.

Answer: B. Explanation: Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Stage-3 boundary without changing its institution, unit or status?

A. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. B. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. C. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. D. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct.

Answer: C. Explanation: Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. The other options change the scientific category, institutional owner, measurement, mission rung or operating

status.

Q40. Which option avoids the standard UPSC close-option trap about Stage-3 boundary?

A. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. B. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. C. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. D. Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium.

Answer: D. Explanation: Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Thorium-U233 boundary?

A. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. B. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. C. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. D. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop.

Answer: A. Explanation: Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Thorium-U233 boundary?

A. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. B. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. C. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. D. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct.

Answer: B. Explanation: Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Thorium-U233 boundary without changing its institution, unit or status?

A. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. B. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. C. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. D. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages.

Answer: C. Explanation: Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Thorium-U233 boundary?

A. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. B. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. C. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. D. Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints.

Answer: D. Explanation: Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Closed-cycle boundary?

A. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. B. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. C. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. D. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR.

Answer: A. Explanation: A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Closed-cycle boundary?

A. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. B. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. C. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. D. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages.

Answer: B. Explanation: A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Closed-cycle boundary without changing its institution, unit or status?

A. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. B. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. C. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. D. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant.

Answer: C. Explanation: A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Closed-cycle boundary?

A. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. B. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. C. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. D. A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop.

Answer: D. Explanation: A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies DAE-BARC boundary?

A. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. B. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. C. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. D. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages.

Answer: A. Explanation: DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of DAE-BARC boundary?

A. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. B. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. C. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. D. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages.

Answer: B. Explanation: DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses DAE-BARC boundary without changing its institution, unit or status?

A. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. B. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. C. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. D. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones.

Answer: C. Explanation: DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. The

other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about DAE-BARC boundary?

A. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. B. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. C. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. D. DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct.

Answer: D. Explanation: DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Operator-regulator boundary?

A. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. B. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. C. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. D. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant.

Answer: A. Explanation: NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Operator-regulator boundary?

A. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. B. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. C. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. D. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant.

Answer: B. Explanation: NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Operator-regulator boundary without changing its institution, unit or status?

A. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. B. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. C. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. D. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect.

Answer: C. Explanation: NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Operator-regulator boundary?

A. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. B. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. C. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. D. NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged.

Answer: D. Explanation: NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies IGCAR-BHAVINI boundary?

A. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. B. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. C. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. D. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages.

Answer: A. Explanation: IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of IGCAR-BHAVINI boundary?

A. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. B. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. C. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. D. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect.

Answer: B. Explanation: IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses IGCAR-BHAVINI boundary without changing its institution, unit or status?

A. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. B. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. C. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. D. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect.

Answer: C. Explanation: IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about IGCAR-BHAVINI boundary?

A. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. B. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. C. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. D. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR.

Answer: D. Explanation: IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Fuel-chain boundary?

A. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. B. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. C. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. D. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones.

Answer: A. Explanation: UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Fuel-chain boundary?

A. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. B. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. C. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. D. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone.

Answer: B. Explanation: UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Fuel-chain boundary without changing its institution, unit or status?

A. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. B. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. C. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. D. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone.

Answer: C. Explanation: UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Fuel-chain boundary?

A. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. B. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. C. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. D. UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages.

Answer: D. Explanation: UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Reactor-type boundary?

A. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. B. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. C. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. D. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone.

Answer: A. Explanation: PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Reactor-type boundary?

A. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. B. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. C. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. D. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect.

Answer: B. Explanation: PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Reactor-type boundary without changing its institution, unit or status?

A. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. B. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. C. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. D. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation.

Answer: C. Explanation: PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. The

other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Reactor-type boundary?

A. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. B. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. C. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. D. PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant.

Answer: D. Explanation: PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Capacity-generation boundary?

A. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. B. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. C. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. D. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect.

Answer: A. Explanation: Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Capacity-generation boundary?

A. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. B. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. C. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. D. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation.

Answer: B. Explanation: Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Capacity-generation boundary without changing its institution, unit or status?

A. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. B. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. C. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. D. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation.

Answer: C. Explanation: Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Capacity-generation boundary?

A. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. B. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. C. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. D. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones.

Answer: D. Explanation: Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Legal-status boundary?

A. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. B. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. C. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. D. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation.

Answer: A. Explanation: Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Legal-status boundary?

A. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. B. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. C. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. D. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation.

Answer: B. Explanation: Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Legal-status boundary without changing its institution, unit or status?

A. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. B. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. C. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. D. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation.

Answer: C. Explanation: Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. The other options change the scientific category, institutional owner, measurement, mission rung

or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Legal-status boundary?

A. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. B. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. C. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. D. Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect.

Answer: D. Explanation: Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile nuclear boundary?

A. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. B. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. C. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. D. Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation.

Answer: A. Explanation: Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile nuclear boundary?

A. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. B. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. C. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. D. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast.

Answer: B. Explanation: Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile nuclear boundary without changing its institution, unit or status?

A. Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. B. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. C. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. D. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel.

Answer: C. Explanation: Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile nuclear boundary?

A. U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. B. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. C. Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. D. Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone.

Answer: D. Explanation: Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route a 2020 IAEA-safeguards objective concept here. The package adds original practice but invents no direct Mains PYQ or objective key.

PYQ DEMAND CARD 1 - 2020 Prelims GS-I

Demand: IAEA safeguards on domestic and imported nuclear reactors.

Status: Verified routed objective concept; no unavailable answer key is inferred.

Model solution: DAE-BARC boundary: DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. **Operator-regulator boundary:** NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. **Legal-status boundary:** Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. **Volatile nuclear boundary:** Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 1 - 2020 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: DAE-BARC boundary: DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. **Operator-regulator boundary:** NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. **Legal-status boundary:** Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. **Volatile nuclear boundary:** Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: IAEA safeguards on domestic and imported nuclear reactors. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed objective concept; no unavailable answer key is inferred. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: DAE-BARC boundary: DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. **Operator-regulator boundary:** NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. **Legal-status boundary:** Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. **Volatile nuclear boundary:** Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2020 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish first criticality, grid synchronisation, installed capacity and generation. Answer in about 150 words.

Model thesis: Claim: Fission-power chain. **Named evidence/example:** Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Criticality boundary. **Named evidence/example:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PFBR-status boundary. **Named evidence/example:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation.
- Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation.
- The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation.
- Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones.

Qualified conclusion: Claim: Fission-power chain. **Named evidence/example:** Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Criticality boundary. **Named evidence/example:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PFBR-status boundary. **Named evidence/example:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **answer** requires a direct position on "Distinguish first criticality, grid synchronisation, installed capacity and generation....", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Fission-power chain. **Named evidence/example:** Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Criticality boundary. **Named evidence/example:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PFBR-status boundary. **Named evidence/example:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Fission-power chain. **Named evidence/example:** Nuclear fission splits a heavy nucleus, releases heat and neutrons, transfers heat through coolant, makes steam and drives a turbine-generator; installed reactor capacity is not electrical generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Criticality boundary. **Named evidence/example:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PFBR-status boundary. **Named evidence/example:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Distinguish first criticality, grid synchronisation, installed capacity and generation....", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain the fissile-fertile distinction in India's three-stage programme. Answer in about 150 words.

Model thesis: Claim: Fissile-fertile boundary. **Named evidence/example:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-3 boundary. **Named evidence/example:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thorium-U233 boundary. **Named evidence/example:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel.
- Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy.
- Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.
- Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium.
- Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints.

Qualified conclusion: Claim: Fissile-fertile boundary. **Named evidence/example:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-3 boundary. **Named evidence/example:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thorium-U233 boundary. **Named evidence/example:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain the fissile-fertile distinction in India's three-stage programme. Answer in about 150...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Fissile-fertile boundary. **Named evidence/example:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-3 boundary. **Named evidence/example:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thorium-U233 boundary. **Named evidence/example:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic

consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Fissile-fertile boundary. **Named evidence/example:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-3 boundary. **Named evidence/example:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thorium-U233 boundary. **Named evidence/example:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain the fissile-fertile distinction in India's three-stage programme. Answer in about 150...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain how Stage 1 creates the material bridge to Stage 2. Answer in about 250 words.

Model thesis: Claim: Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Plutonium-stream boundary. **Named evidence/example:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Closed-cycle boundary. **Named evidence/example:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy.
- Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states.
- Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.
- A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop.

Qualified conclusion: Claim: Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Plutonium-stream boundary. **Named evidence/example:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Closed-cycle boundary. **Named evidence/example:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain how Stage 1 creates the material bridge to Stage 2. Answer in about 250 words.", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Plutonium-stream boundary. **Named evidence/example:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Closed-cycle boundary. **Named evidence/example:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Plutonium-stream boundary. **Named evidence/example:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Closed-cycle boundary. **Named evidence/example:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain how Stage 1 creates the material bridge to Stage 2. Answer in about 250 words.", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess PFBR as a Stage-2 milestone without overstating deployment. Answer in about 250 words.

Model thesis: Claim: Criticality boundary. **Named evidence/example:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thermal-fast boundary. **Named evidence/example:** Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PFBR-status boundary. **Named evidence/example:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IGCAR-BHAVINI boundary. **Named evidence/example:** IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation.
- Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast.
- Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.
- The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation.
- IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR.
- Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones.

Qualified conclusion: Claim: Criticality boundary. **Named evidence/example:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thermal-fast boundary. **Named evidence/example:** Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PFBR-status boundary. **Named evidence/example:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IGCAR-BHAVINI boundary. **Named evidence/example:** IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **assess** requires a direct position on "Assess PFBR as a Stage-2 milestone without overstating deployment. Answer in about 250 words.", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Criticality boundary. **Named evidence/example:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thermal-fast boundary. **Named evidence/example:** Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PFBR-status boundary. **Named evidence/example:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IGCAR-BHAVINI boundary. **Named evidence/example:** IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit,

source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

5. **Claim and named evidence:** IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Criticality boundary. **Named evidence/example:** Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thermal-fast boundary. **Named evidence/example:** Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PFBR-status boundary. **Named evidence/example:** The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IGCAR-BHAVINI boundary. **Named evidence/example:** IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Assess PFBR as a Stage-2 milestone without overstating deployment. Answer in about 250 words.", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate India's three-stage programme as a closed fuel-cycle strategy. Answer in about 300 words.

Model thesis: Claim: Fissile-fertile boundary. **Named evidence/example:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Plutonium-stream boundary. **Named evidence/example:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-3 boundary. **Named evidence/example:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thorium-U233 boundary. **Named evidence/example:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Closed-cycle boundary. **Named evidence/example:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Fuel-chain boundary. **Named evidence/example:** UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel.
- Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy.
- Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states.
- Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet.

- Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium.
- Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints.
- A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop.
- UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages.

Qualified conclusion: Claim: Fissile-fertile boundary. **Named evidence/example:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Plutonium-stream boundary. **Named evidence/example:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-3 boundary. **Named evidence/example:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thorium-U233 boundary. **Named evidence/example:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Closed-cycle boundary. **Named evidence/example:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Fuel-chain boundary. **Named evidence/example:** UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **evaluate** requires a direct position on "Evaluate India's three-stage programme as a closed fuel-cycle strategy. Answer in about 300...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Fissile-fertile boundary. **Named evidence/example:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Plutonium-stream boundary. **Named evidence/example:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-3 boundary. **Named evidence/example:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thorium-U233 boundary. **Named evidence/example:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Closed-cycle boundary. **Named evidence/example:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Fuel-chain boundary. **Named evidence/example:** UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Fissile-fertile boundary. **Named evidence/example:** U-235, Pu-239 and U-233 are fissile, while U-238 and Th-232 are fertile materials that can breed fissile nuclides; fertile material is not ready-to-use fissile fuel. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-1 boundary. **Named evidence/example:** Stage 1 centres on Pressurised Heavy Water Reactors using natural uranium and heavy water, producing electricity and plutonium-bearing spent fuel for the closed-cycle strategy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Plutonium-stream boundary. **Named evidence/example:** Plutonium used for the breeder bridge is recovered from irradiated Stage-1 fuel through reprocessing; spent fuel, separated fissile material and fresh reactor fuel are distinct material states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-2 boundary. **Named evidence/example:** Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Stage-3 boundary. **Named evidence/example:** Stage 3 is the long-term thorium-U-233 utilisation horizon; it depends on prior fissile inventory, irradiation, reprocessing and fuel fabrication rather than direct burning of mined thorium. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thorium-U233 boundary. **Named evidence/example:** Th-232 absorbs a neutron and ultimately yields fissile U-233, while U-232 contamination complicates handling through intense gamma-emitting decay products; thorium abundance does not remove engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Closed-cycle boundary. **Named evidence/example:** A closed fuel cycle links irradiation, cooling, reprocessing, remote fuel fabrication, reactor reuse and waste management; breeder scaling depends on the whole material loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Fuel-chain boundary. **Named evidence/example:** UCIL mines and mills uranium, NFC fabricates fuel and reactor components, and IREL processes monazite associated with thorium and rare earths; resource, fabrication and reactor operation are separate stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Evaluate India's three-stage programme as a closed fuel-cycle strategy. Answer in about 300...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Analyse nuclear expansion through reactor choice, institutions, regulation and legal-status discipline. Answer in about 300 words.

Model thesis: Claim: Control-safety boundary. **Named evidence/example:** Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thermal-fast boundary. **Named evidence/example:** Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DAE-BARC boundary. **Named evidence/example:** DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Operator-regulator boundary. **Named evidence/example:** NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IGCAR-BHAVINI boundary. **Named evidence/example:** IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Reactor-type boundary. **Named evidence/example:** PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Legal-status boundary. **Named evidence/example:** Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile nuclear boundary. **Named evidence/example:** Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable.
- Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast.
- DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct.
- NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged.
- IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR.
- PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant.
- Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones.
- Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect.
- Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone.

Qualified conclusion: Claim: Control-safety boundary. **Named evidence/example:** Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thermal-fast boundary. **Named evidence/example:** Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DAE-BARC boundary. **Named evidence/example:** DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Operator-regulator boundary. **Named evidence/example:** NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IGCAR-BHAVINI boundary. **Named evidence/example:** IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Reactor-type boundary. **Named evidence/example:** PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Legal-status boundary. **Named evidence/example:** Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile nuclear boundary. **Named evidence/example:** Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **analyse** requires a direct position on "Analyse nuclear expansion through reactor choice, institutions, regulation and legal-status...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Control-safety boundary. **Named evidence/example:** Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Thermal-fast boundary. **Named evidence/example:** Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DAE-BARC boundary. **Named evidence/example:** DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Operator-regulator boundary. **Named evidence/example:** NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IGCAR-BHAVINI boundary. **Named evidence/example:** IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Reactor-type boundary. **Named evidence/example:** PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capacity-generation boundary. **Named evidence/example:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Legal-status boundary. **Named evidence/example:** Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile nuclear boundary. **Named evidence/example:** Fleet size, capacity, generation, fuel inventory, reactor status, legal commencement, cost and schedule require dated DAE, NPCIL, AERB or statutory evidence and must not be inferred from a milestone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Control rods absorb neutrons to regulate the reaction, a moderator slows neutrons in thermal reactors, coolant removes heat, and containment limits release; these functions are not interchangeable. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** DAE provides the governmental policy and strategic umbrella, while BARC performs reactor, fuel-cycle and advanced-concept R&D; policy direction and laboratory execution are distinct. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** NPCIL develops and operates commercial civilian nuclear plants, while AERB performs safety review and regulatory oversight; operator and regulator must never be merged. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** PHWR, PWR, BWR, FBR, AHWR and small-reactor concepts differ in moderator, coolant, fuel, neutron spectrum and development status; a design concept is not an operating plant. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

7. **Claim and named evidence:** Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** Atomic-energy ownership, civil liability and safety regulation are different legal domains; an enacted replacement framework does not repeal existing law until its commencement provisions take effect. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
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Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Analyse nuclear expansion through reactor choice, institutions, regulation and legal-status...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.